



OPEN Obstructive sleep apnea and genotype rs6843082 as a risk factor for cerebrovascular accident

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NO previous studies have examined the simultaneous effects of obstructive sleep apnea (OSA), hypertension, and the SNP rs6843082 on stroke. We aimed to explore whether these elements together, play a role as risk factors for stroke. Data was obtained from the Taiwan Biobank and the National Health Insurance database. We used logistic regression analysis to investigate the effect of OSA and hypertension as a risk factor for stroke in different genotypes. We found that OSA and hypertension was associated with stroke in those with the rs6843082 genotype. People with OSA and hypertension together with the rs6843082 genotype (GA + AA) showed a statistically significant difference as a risk for stroke (OR, 2.57; 95% CI, 1.53 to 4.33). However, there was no statistically significant difference in those people with OSA but without hypertension (OR, 0.53; 95% CI, 0.13 to 2.25). After further stratification by combination of OSA and hypertension, those with genotype rs6843082 (GG) had higher risk odds than those with OSA and those with hypertension alone (OR, 5.46, 95% CI, 3.46 to 8.60). Individuals with genotype rs6843082 (GA + AA), OSA and hypertension together had the highest risk for stroke (OR, 6.25, 95% CI, 3.63 to 10.76) and those with OSA and no hypertension (OR, 0.57, 95% CI, 0.14 to 2.36) had no significant risk. Our findings showed that people with genotype rs6843082 (GG), with or without hypertension had OSA as a risk factor for stroke. For individuals with the genotype rs6843082 (GA + AA), those with hypertension, OSA is a risk factor for stroke, and for those without hypertension, OSA is not associated with stroke.

Keywords Stroke, Obstructive sleep apnea (OSA), Hypertension, Single nucleotide polymorphism, rs6843082

Abbreviations

OSA Obstructive sleep apnea
SNP Single nucleotide polymorphism
CVD Cardiovascular disease

Stroke remains the second-leading cause of death and the third-leading cause of death and disability combined in the world¹. Research data support a greater focus on stroke prevention and treatment in low-income countries, where there was a much greater burden of stroke in terms of mortality and disability².

The significant burden of stroke on the individual and society is undisputed³.

Stroke risk is evaluated on an individual and societal basis based on the presence of modifiable (e.g., hypertension, atrial fibrillation) and nonmodifiable risk factors (e.g., male sex, age and ethnicity)⁴.

The significant risk factors for all strokes in a previous study of 3000 patients were: history of hypertension; current smoking; waist-to-hip ratio; diet risk score; regular physical activity; diabetes mellitus; alcohol intake;

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psychosocial stress and depression; cardiac causes; and ratio of apolipoproteins B to A1⁵. Hypertension is clearly one of the common and important risk factors for stroke.

Arzt et al. study data demonstrate a strong association between moderate to severe sleep-disordered breathing and prevalent stroke, independent of confounding factors⁶.

Sleep disorders are highly prevalent in patients at risk for stroke, and may be modifiable risk factors for stroke⁷.

The study by Yaggi et al. indicates that obstructive sleep apnea (OSA) significantly increases the risk of stroke or death from any cause. After adjustment for age, sex, race, smoking status, alcohol-consumption status, body-mass index, and the presence or absence of diabetes mellitus, hyperlipidemia, atrial fibrillation, and hypertension, the obstructive sleep apnea syndrome retained a statistically significant association with stroke or death (hazard ratio, 1.97; 95% confidence interval, 1.12 to 3.48; $P = 0.01$)⁸.

The results highlight the independent impact of OSA on stroke risk.

Yaranov et al. also observed a positive dose effect of the apnea-hypopnea index on the rate of stroke, and the results demonstrated that OSA is associated with an increased incidence of stroke, and this association remained significant after accounting for other cardiovascular risk factors⁹.

This relationship between OSA and stroke was confirmed in a recent meta-analysis of 12 prospective cohort studies by Wang et al. that encompassed 25,760 subjects whereby the relative risk of incident fatal and non-fatal stroke for severe OSA compared to SA was 2.15 (95% confidence interval, 1.42–3.24)¹⁰.

Furthermore, their study showed that severe OSA is significantly and independently associated with an increased risk of cardiovascular disease (CVD), stroke, and all-cause mortality¹⁰. Several prospective cohort studies have identified OSA as an independent risk factor for stroke after accounting for potential confounding risk factors^{7,11,12}.

It can be seen that OSA should be highlighted as a risk factor for stroke.

Stroke is a multifactorial disease with genetic and environmental components, in which gene–gene and gene–environment interactions are important¹³. In the other cohort study, the PITX2 and ZFHX3 genes are related to cardioembolic stroke¹⁴. This study indicated that there was a significant association between the PITX2 gene and stroke risk¹⁵. Other studies have also verified that two single nucleotide polymorphisms (SNPs) rs6843082 (PITX2, 4q25) and rs7193343 (ZFHX3, 16q22) are common loci in many stroke genetic studies¹⁶.

However, these studies examined the association of these SNPs with stroke development, but in these genetic locus analyses, no association was indicated between hypertension, OSA, and stroke SNPs. Furthermore, previous research has shown that both OSA and hypertension are independent risk factors for stroke. Therefore, this study aims to investigate the association between OSA, hypertension, and stroke genes.

Materials and methods

Data source

Data for this retrospective cohort study were obtained from the following two sources: (1) the National Health Insurance Research Database (NHIRD, with data to 1998–2015) to identify stroke, OSA, hypertension, diabetes and Atrial Fibrillation Disease) and (2) the Taiwan Biobank (TWB), a national health resource containing basic demographic information, personal health activities, physical examination and genotype data to 2008–2015. These databases were linked and personal identification numbers (PIN) were used to obtain related stroke information. The data used in this study were obtained from the Taiwan Biobank dataset, and informed consent was obtained from all individual participants in the study. This study was approved by the Institutional Review Board of Chung Shan Medical University Hospital (CS1-20009) approved this study in this context.

Patient identification

Our initial recruitment included 17,985 participants from the Taiwan Biobank. Demographic information (sex, age, body mass index [BMI]), lifestyle (regular exercise, smoking and alcohol consumption) and genotypic data were retrieved from the biobank database. Patients with incomplete questionnaires ($n = 56$) and genotyping information ($n = 14$) were excluded. Finally, this research includes the contrast of 1020 stroke patients and 16,895 control patients. Alcohol drinkers were defined as persons who had consumed more than 150 ml of alcohol per week six months prior to the health examination. Physical activity included any amount of exercise activity at lasted for 3 times a week and lasting at least 30 min each time. Smokers were defined as smoked for more than six months. A second-hand smoker was defined as someone who was currently exposed to the smoke of other people for at least 5 min per day.

Genetic variant selection

A Chromosome 4q25 gene PITX2 SNP(rs6843082) and a Chromosome 16q22 gene ZFHX3 SNP (rs7193343) were selected based on a literature search associated with cardioembolic stroke and genotyping was performed using custom Taiwan biobank chips. SNP genotyping was carried out using custom Taiwan Biobank chips and ran on the Axiom™ Genome-Wide Array Plate System (Affymetrix, Santa Clara, CA, USA). For the TWB SNPs array, we followed a standard quality control procedure to exclude a low recall rate ($< 95\%$), p for the Hardy-Weinberg equilibrium test of $< 1.0 \times 10^{-3}$, and a minor allele frequency of < 0.05 . Multivariate unconditional logistic regression was used, comparing GG or GA with AA for rs6843082 and TT and TC with CC for rs7193343.

Definition of outcomes

We identified patients diagnosed with stroke (ICD-9-CM: 430–432, 433–437), OSA (ICD-9-CM: 327.23, 780.51, 780.53, 780.57), hypertension (ICD-9-CM: 401–405), diabetes mellitus (ICD-9-CM: 250), hyperlipidemia (ICD-9-CM: 272) and atrial fibrillation (ICD-9-CM: 427.3) using two outpatient visits or admission to NHIRD from

1998 to 2015, and the first diagnosis date before acceptance date. Body mass index (BMI) was calculated as weight in kilograms divided by height in meters squared (kg/m2).

Statistical analysis

The chi-square test and the t-test were used to compare the differences between discrete and continuous variables. Logistic regression analysis was used to investigate the effects of OSA and hypertension on stroke in the difference genotype. The odds ratios (OR) were estimated with their 95% confidence intervals. Statistical significance was established with a value of $p < 0.05$. Statistical analyzes were performed using the statistical analysis system (SAS) (version 9.4) and PLINK.

Result

According to the research conditions and definitions, the study consisted of 1,020 stroke patients and 16,895 control patients.

Among the study participants listed in Table 1, the risk factors for stroke were OSA (OR, 2.182; 95% CI: 1.634 to 2.914), age (OR, 3.032; 95% CI: 2.502 to 3.676), diabetes (OR, 1.521; 95% CI: 1.303 to 1.776), hypertension (OR, 2.443; 95% CI: 2.096 to 2.848), hyperlipidemia (OR, 1.669; 95% CI: 1.429 to 1.948), atrial fibrillation (OR, 3.089; 95% CI: 1.952 to 4.889).

Subject characteristics

Table 2 lists the study participants, among stroke patients with the rs6843082 (GG) genotype, there were 587 (5.85%), and 221 (2.2%) were diagnosed with OSA disease. In the genotype rs6843082 (GA + AA), there were 433 (5.5%) stroke patients and 186 (2.36%) were diagnosed with OSA disease. In addition, there are statistically significant differences in education ($p = 0.022$) and atrial fibrillation ($p = 0.018$).

	OR	95%CI	
OSA (ref: No)			
Yes	2.182	1.634	2.914
LOC729065-rs6843082 (ref: GG)			
GA	0.987	0.859	1.134
AA	0.784	0.585	1.051
ZFHX3-rs7193343 (ref: TT)			
TC	0.991	0.862	1.139
CC	0.958	0.764	1.202
Sex (ref: Female)			
Male	1.062	0.904	1.248
Age (ref:30–50)			
50–70	3.032	2.502	3.676
Education (ref: Elementary school)			
Junior and Senior high school	0.904	0.727	1.124
University above	0.739	0.588	0.929
Smoke (ref: No)			
Yes	0.912	0.762	1.091
Alcohol drink (ref: No)			
Yes	1.207	0.975	1.494
Second-hand smoke (ref: No)			
Yes	0.864	0.682	1.093
Physical activity (ref: No)			
Yes	1.119	0.975	1.286
BMI	0.991	0.971	1.011
Diabetes (ref: No)			
Yes	1.521	1.303	1.776
Hypertension (ref: No)			
Yes	2.443	2.096	2.848
Hyperlipidemia (ref: No)			
Yes	1.669	1.429	1.948
Atrial fibrillation (ref: No)			
Yes	3.089	1.952	4.889

Table 1. Stroke risk factors.

	rs6843082(GG)		rs6843082(GA + AA)		P value
	N	%	N	%	
Stroke					0.316
No	9452	94.15	7443	94.5	
Yes	587	5.85	433	5.5	
OSA					0.475
No	9818	97.8	7690	97.64	
Yes	221	2.2	186	2.36	
rs7193343 (ZFHX3)					0.099
TT	4602	45.84	3513	44.6	
TC + CC	5437	54.16	4363	55.4	
Sex					0.474
Female	5196	51.76	4034	51.22	
Male	4843	48.24	3842	48.78	
Age					0.244
30–50	5189	51.69	4140	52.56	
50–70	4850	48.31	3736	47.44	
Education					0.022
Elementary school	621	6.19	414	5.26	
Junior and Senior high school	3922	39.07	3061	38.86	
University above	5496	54.75	4401	55.88	
Smoke					0.566
No	7587	75.58	5923	75.2	
Yes	2452	24.42	1953	24.8	
Drink					0.304
No	9012	89.77	7033	89.3	
Yes	1027	10.23	843	10.7	
Second-hand smoke					0.569
No	8854	88.2	6968	88.47	
Yes	1185	11.8	908	11.53	
Physical activity					0.867
No	5884	58.61	4626	58.74	
Yes	4155	41.39	3250	41.26	
Diabetes					0.679
No	8680	86.46	6793	86.25	
Yes	1359	13.54	1083	13.75	
Hypertension					0.703
No	7668	76.38	6035	76.63	
Yes	2371	23.62	1841	23.37	
Hyperlipidemia					0.957
No	7004	69.77	5492	69.73	
Yes	3035	30.23	2384	30.27	
Atrial fibrillation					0.018
No	9970	99.31	7843	99.58	
Yes	69	0.69	33	0.42	
BMI(Mean, S.D)	24.24	3.61	24.31	3.66	0.219

Table 2. Descriptive data of the participants.**Effect of genotype**

Table 3, stratified by genotype, analyses the relationship between people with different genotypes, stroke patients, OSA, and hypertension.

OSA disease (OR, 2.31; 95% CI, 1.60 to 3.36) with hypertension (OR, 2.46; 95% CI, 2.01 to 3.02), were both statistically significant as a risk factor compared to the control group in patients with the genotype rs6843082(GG). OSA disease (OR, 1.98; 95% CI, 1.24 to 3.16) with hypertension (OR, 2.42; 95% CI, 1.92 to 3.07), both showed statistical significance when compared to the control group in patients with Genotype rs6843082(GA + AA).

	rs6843082(GG)		rs6843082(GA + AA)	
	OR	95% CI	OR	95% CI
OSA (ref: No)				
Yes	2.31	1.60 - 3.36	1.98	1.24 - 3.16
Hypertension (ref: No)				
Yes	2.46	2.01 - 3.02	2.42	1.92 - 3.07
ZFHX3-rs7193343 (ref: TT)				
TC+CC	0.97	0.82 - 1.16	1.00	0.82 - 1.23
Sex (ref: Female)				
Male	1.05	0.84 - 1.29	1.07	0.84 - 1.37
Age (ref:30–50)				
50–70	2.54	1.99 - 3.25	3.97	2.91 - 5.40
Education (ref: Elementary school)				
Junior and Senior high school	0.91	0.69 - 1.20	0.91	0.64 - 1.28
University above	0.70	0.52 - 0.94	0.80	0.56 - 1.15
Smoke (ref: No)				
Yes	0.96	0.76 - 1.22	0.86	0.65 - 1.13
Alcohol drink (ref: No)				
Yes	1.27	0.96 - 1.67	1.12	0.80 - 1.57
Second-hand smoke (ref: No)				
Yes	0.74	0.54 - 1.02	1.04	0.73 - 1.47
Physical activity (ref: No)				
Yes	1.31	1.09 - 1.57	0.91	0.74 - 1.13
BMI	1.02	0.99 - 1.05	0.95	0.92 - 0.98
Diabetes (ref: No)				
Yes	1.37	1.11 - 1.69	1.71	1.35 - 2.16
Hyperlipidemia (ref: No)				
Yes	1.61	1.32 - 1.98	1.76	1.39 - 2.24
Atrial fibrillation (ref: No)				
Yes	2.98	1.70 - 5.22	3.38	1.50 - 7.59

Table 3. Association of stroke with OSA and hypertension by rs6843082.

Effect of hypertension

Additionally, genotype grouping and hypertension stratification were used to explore the relationship between OSA, hypertension, and stroke in people with different genotypes.

Table 4 shows the results of people with genotype rs6843082 (GG) grouped by hypertension, compared to patients without OSA, those with OSA without hypertension (OR, 2.75; 95% CI: 1.24 to 3.16), with OSA with hypertension (OR, 2.17; 95% CI: 1.41 to 3.34), all were statistically significant, suggesting that OSA was associated with stroke. The statistical results showed that there was no interaction between OSA and hypertension ($P = 0.627$).

Table 5 shows the results of people with the genotype rs6843082 (GA + AA) grouped by hypertension. Compared to patients without OSA, there were no statistically significant differences between patients with OSA and without hypertension (OR, 0.53; 95% CI: 0.13 to 2.25), but in those with OSA and hypertension (OR, 2.57; 95% CI: 1.53 to 4.33), there was a statistically significant difference, showing that only those with hypertension had OSA as a risk factor for stroke. The statistical results showed that there was an interaction between OSA and hypertension ($P = 0.0429$).

Subjective outcomes

Table 6 further analyzed the relationship between different genotypes of rs6843082 and stroke based on the combination of OSA and hypertension. The results showed that people with genotype rs6843082(GG), regardless of whether they had no OSA and hypertension (OR = 2.49, 95% CI: 2.02 to 3.06), had OSA and no hypertension (OR, 2.68, 95% CI: 1.35 to 5.33), OSA and hypertension (OR, 5.46, 95% CI: 3.46 to 8.60), showed statistically significant differences from the control group and the risk probability is significantly increased. This indicates that no matter whether there is hypertension or not, OSA was a risk factor for stroke.

The genotype rs6843082 (GA + AA) showed that people without OSA and hypertension (OR, 2.30, 95% CI: 1.81 to 2.92), or with OSA and hypertension (OR, 6.25, 95% CI: 3.63 to 8.60) had a statistically significant difference, but there was no statistically significant difference between those with OSA and those without hypertension (OR, 0.57, 95% CI: 0.14 to 2.36).

	Non-hypertension		Hypertension	
	OR	95% CI	OR	95% CI
OSA (ref: No)				
Yes	2.75	1.36 - 5.57	2.17	1.41 - 3.34
ZFHX3-rs7193343 (ref: TT)				
TC + CC	1.09	0.83 - 1.42	0.88	0.70 - 1.11
Sex (ref: Female)				
Male	0.93	0.67 - 1.28	1.14	0.85 - 1.52
Age (ref:30–50)				
50–70	3.04	2.18 - 4.24	1.59	1.11 - 2.29
Education (ref: Elementary school)				
Junior and Senior high school	0.77	0.49 - 1.21	0.91	0.64 - 1.29
University above	0.50	0.31 - 0.80	0.82	0.57 - 1.20
Smoke (ref: No)				
Yes	0.78	0.53 - 1.16	1.06	0.79 - 1.42
Alcohol drink (ref: No)				
Yes	1.55	0.99 - 2.44	1.13	0.80 - 1.59
Second-hand smoke (ref: No)				
Yes	0.87	0.54 - 1.40	0.68	0.44 - 1.05
Physical activity (ref: No)				
Yes	1.37	1.04 - 1.80	1.22	0.96 - 1.56
BMI	1.03	0.99 - 1.07	1.01	0.98 - 1.04
Diabetes (ref: No)				
Yes	1.49	1.03 - 2.15	1.38	1.07 - 1.76
Hyperlipidemia (ref: No)				
Yes	2.05	1.53 - 2.75	1.23	0.94 - 1.60
Atrial fibrillation (ref: No)				
Yes	3.33	1.09 - 10.18	2.79	1.47 - 5.32

Table 4. Association of stroke with OSA based on the hypertension in rs6843082(GG) genotype.
*OSA*Hypertension, P=0.627

Discussion

The pathogenesis of stroke is complex. Epidemiological studies have identified various risk factors for stroke, including obstructive sleep apnea (OSA), hypertension, dietary risks, diabetes, obesity, smoking, air pollution, alcohol consumption, hypercholesterolemia, physical inactivity, aging, and gender^{7,17}.

Stroke may also be caused by genetic factors; the interaction between genetic and disease factors may lead to stroke in certain individuals. To our knowledge, this is the first cohort study investigating the association between OSA, hypertension, the PITX2 SNP rs6843082, and stroke using data from the Taiwan Biobank and clinical research databases. The combined results from these data sources indicate that different genotypes of SNP rs6843082, in combination with OSA and hypertension, have different impacts on the risk of stroke. Understanding these interactions is crucial for developing targeted prevention strategies and personalized treatments for high-risk stroke populations, as well as for public health implications.

Our findings have several implications for clinical practice. First, we observed that individuals with the rs6843082 genotype (GG) are at an increased risk of stroke when affected by OSA, regardless of their hypertension status, indicating that OSA is a risk factor for stroke. In contrast, individuals with the rs6843082 genotype (GA + AA) and hypertension show an increased risk of stroke when affected by OSA, whereas OSA does not significantly impact the stroke risk in those without hypertension.

Interestingly, our study found that individuals with the rs6843082 genotype (GG) have a higher risk of stroke compared to those with OSA or hypertension alone. This suggests that the presence of the rs6843082 genotype (GG) may synergistically increase the risk of stroke when combined with OSA and hypertension.

Furthermore, our study found that individuals with the rs6843082 genotype (GA + AA) have a significantly higher stroke risk compared to those with the GG genotype. When stratifying by the presence of hypertension and OSA, we found that the combination of hypertension, OSA, and the rs6843082 genotype (GA + AA) significantly increases stroke risk. This indicates a potential gene-environment interaction, where the genetic susceptibility conferred by the rs6843082 SNP amplifies the risk of stroke in the presence of hypertension and OSA.

Stroke is a leading cause of global mortality and morbidity, with hypertension being one of its most significant risk factors^{18,19}. OSA is also considered an independent risk factor for stroke, with this relationship being independent of other risk factors^{8,20}. The direct and profound cardio-respiratory effects of OSA, including

	Non-hypertension		Hypertension	
	OR	95% CI	OR	95% CI
OSA (ref: No)				
Yes	0.53	0.13 - 2.25	2.57	1.53 - 4.33
ZFHX3-rs7193343 (ref: TT)				
TC + CC	0.99	0.72 - 1.34	1.02	0.78 - 1.35
Sex (ref: Female)				
Male	0.78	0.53 - 1.15	1.35	0.97 - 1.88
Age (ref:30–50)				
50–70	4.39	2.95 - 6.52	2.89	1.76 - 4.72
Education (ref: Elementary school)				
Junior and Senior high school	0.83	0.46 - 1.48	0.94	0.61 - 1.45
University above	0.82	0.46 - 1.49	0.77	0.48 - 1.22
Smoke (ref: No)				
Yes	1.10	0.69 - 1.75	0.73	0.52 - 1.04
Alcohol drink (ref: No)				
Yes	1.02	0.57 - 1.84	1.13	0.75 - 1.72
Second-hand smoke (ref: No)				
Yes	0.85	0.48 - 1.50	1.20	0.76 - 1.88
Physical activity (ref: No)				
Yes	1.00	0.73 - 1.38	0.84	0.63 - 1.12
BMI	0.94	0.89 - 0.99	0.96	0.92 - 1.00
Diabetes (ref: No)				
Yes	2.53	1.71 - 3.73	1.45	1.09 - 1.93
Hyperlipidemia (ref: No)				
Yes	2.07	1.47 - 2.91	1.43	1.04 - 1.98
Atrial fibrillation (ref: No)				
Yes	-		4.40	1.86 - 10.37

Table 5. Association of stroke with OSA based on the hypertension in rs6843082(GA + AA) genotype. *OSA*Hypertension, P=0.0429

hypoxemia, changes in heart rate and blood pressure, altered cerebral or other regional blood flow, intrathoracic pressure fluctuations, and increased sympathetic activity, provide multiple potential pathophysiological mechanisms leading to cardiovascular dysfunction^{21,22}.

OSA is common among hypertensive patients, and these conditions often coexist. Hypertension and OSA share common pathophysiological mechanisms, such as endothelial dysfunction, inflammation, and sympathetic overactivity, which synergistically increase the risk of stroke^{23,24}. Additionally, hypertension and OSA are related to similar risk factors for stroke, including obesity, metabolic syndrome, and insulin resistance²⁵.

Several studies have shown a potential association between genetic variations in the PITX2 gene and the risk of stroke. PITX2 is a transcription factor involved in the development of the heart and brain, with previous research identifying SNPs related to cardioembolic stroke, including PITX2 rs6843082 on 4q25 and ZFHX3 rs879324 on 16q22^{26,27}.

In individuals with the rs6843082 genotype (GG), OSA appears to be a risk factor for stroke, independent of hypertension status. Previous studies have found that the presence of the G allele may increase the risk of stroke. The potential mechanisms may involve the exacerbation of arrhythmias, endothelial dysfunction, and cardiovascular complications due to intermittent hypoxia and sympathetic activation associated with OSA^{16,28}. Conversely, in individuals with the rs6843082 genotype (GA + AA) and hypertension, the combination of OSA and hypertension synergistically increases the risk of stroke. However, individuals with OSA alone and without hypertension exhibit a lower stroke risk, suggesting that the presence of the A allele may alter stroke risk. The rs6843082 (GA + AA) genotype may help maintain intracellular redox balance, thereby reducing the risk of stroke and other cardiovascular diseases²⁹. However, more research is needed to further elucidate the specific roles and mechanisms of these genetic variations.

The results also indicate that individuals with the rs6843082 (GA + AA) genotype and OSA should pay more attention to the prevention of hypertension, as the combination of hypertension significantly increases the risk of stroke by 6.25 times compared to those without OSA and without hypertension.

Additionally, our study found that individuals with the rs6843082 genotype (GA + AA) have a significantly higher stroke risk compared to those with the GG genotype. This may be attributed to a more pronounced adverse effect on vascular function and a higher burden of comorbidities. This suggests a potential gene-environment

	rs6843082(GG)		rs6843082(GA + AA)	
	OR	95%CI	OR	95%CI
OSA and Hypertension (ref: Non-OSA, Non-hypertension)				
Non-OSA, hypertension	2.49	2.02 - 3.06	2.30	1.81 - 2.92
OSA, Non-hypertension	2.68	1.35 - 5.33	0.57	0.14 - 2.36
OSA, Hypertension	5.46	3.46 - 8.60	6.25	3.63 - 10.76
ZFHX3-rs7193343 (ref: TT)				
TC + CC	0.97	0.82 - 1.16	1.00	0.81 - 1.22
Sex (ref: Female)				
Male	1.04	0.84 - 1.29	1.08	0.84 - 1.38
Age (ref:30–50)				
50–70	2.54	1.99 - 3.25	4.00	2.93 - 5.44
Education (ref: Elementary school)				
Junior and Senior high school	0.91	0.69 - 1.20	0.90	0.64 - 1.28
University above	0.70	0.52 - 0.94	0.80	0.56 - 1.15
Smoke (ref: No)				
Yes	0.96	0.76 - 1.22	0.85	0.64 - 1.12
Alcohol drink (ref: No)				
Yes	1.27	0.96 - 1.67	1.11	0.79 - 1.56
Second-hand smoke (ref: No)				
Yes	0.74	0.54 - 1.02	1.04	0.73 - 1.48
Physical activity (ref: No)				
Yes	1.31	1.09 - 1.57	0.91	0.73 - 1.12
BMI	1.02	0.99 - 1.05	0.95	0.92 - 0.98
Diabetes (ref: No)				
Yes	1.37	1.11 - 1.68	1.73	1.37 - 2.19
Hyperlipidemia (ref: No)				
Yes	1.61	1.32 - 1.97	1.76	1.39 - 2.24
Atrial fibrillation (ref: No)				
Yes	2.96	1.68 - 5.19	3.36	1.49 - 7.55

Table 6. Odds ratios (95% CI) for stroke after combining OSA and hypertension in the different genotype (rs6843082).

interaction, where the genetic susceptibility conferred by the rs6843082 SNP amplifies the risk of stroke in the presence of hypertension and OSA³⁰.

In summary, obstructive sleep apnea and hypertension are significantly associated with an increased risk of stroke, independent of other risk factors. The severity of OSA is linked to an increased risk of this composite outcome.

Our findings show an association between OSA and the PITX2 gene with the risk of stroke. Recent studies have shown that severe obstructive sleep apnea increases the risk of fatal and non-fatal cardiovascular events³¹. Studies indicate that OSA independently increases stroke risk, with an unadjusted relative risk of 5.16 (95% CI: 3.72–6.60)¹¹. In the study population of Yaranov et al., OSA was significantly associated with stroke in multivariate logistic regression analysis, with an adjusted odds ratio of 3.65 (95% CI: 1.252 to 10.623), showing that OSA patients have a risk of stroke⁹. Another GWAS analysis reported that the rs6843082 variant near PITX2 showed a strong association with severe cardiac embolic stroke risk ($p = 2.8 \times 10^{-16}$)¹⁶. Moreover, our study also shows a statistically significant association between the PITX2 SNP rs6843082, OSA, hypertension, and stroke, consistent with previous studies showing this association between PITX2 and ischemic stroke^{32,33}. Therefore, the PITX2 gene may be involved in the circulatory system to some extent and may act on stroke-related risk factors. We speculate that the rs6843082 SNP and PITX2 are directly related to the expression of OSA and hypertension²⁶.

Overall, our findings about the increased risk of stroke in patients with obstructive sleep apnea are consistent with previous reports.

However, in an observational cohort study by Yaggi HK et al., obstructive sleep apnea syndrome significantly increased the risk of stroke or death from any cause, and this increase was independent of other risk factors, including hypertension. Conversely, our study found that different genotypes and the presence of hypertension have different impacts on stroke risk⁸.

Finally, the data analysis shows that SNP 7,193,343 does not show a statistically significant association with OSA, hypertension, and stroke, and no statistical difference was observed after adjusting for relevant factors.

Despite the important findings, some limitations should be acknowledged. First, the study relies on retrospective data from the Taiwan Biobank and the National Health Insurance database, and due to the sample size of stroke patients, this study did not specifically investigate stroke subtypes. Secondly, considering the sample size distribution of the rs6843082 allele, according to SNPedia, the allele distribution of rs6843082 among the Han Chinese is (A; A) 10.2%, (A; G) 42.3%, (G; G) 47.4%^{34,35}. Due to the smaller number of individuals with the rs6843082 A allele in our data, we combined the SNP rs6843082 (G; A) and (A; A) for observation.

Additionally, our analysis considered all known major risk factors for stroke. Although we attempted to control for major cardiovascular risk factors, residual confounding factors may still influence our adjusted risk ratios.

In addition, The AUC values, likelihood ratio chi-square values, and p-values for each tables are all shown in Supplementary Table 1. In this study, although there is a significant disparity in the number of cases and controls, we employed cross-validation to assess the robustness and reliability of our analysis. We randomly split both cases and controls into ten equal parts and conducted the analysis by removing one part at a time, utilizing the remaining nine parts for the analysis. The results are presented in Supplementary Table 2.

In the cross-validation, the OR values ranged from a minimum of 2.031 to a maximum of 2.396, each achieving statistical significance (p value < 0.0001). These results are similar to those obtained using the complete dataset, with an OR value of 2.180.

Finally, the analysis is limited to Taiwanese individuals, and the results may not be generalizable to other populations with different genetic backgrounds and environmental exposures.

The genetics of multifactorial diseases like stroke remain a complex field, and research into new or emerging risk factors remains an active area of investigation. Therefore, variations in the PITX2 gene can be used to identify the risk of stroke in patients. Timely screening for OSA, hypertension, and genetic testing can help individuals better prevent the occurrence of stroke, thereby reducing its impact on individuals and society³⁶.

Conclusion

Our findings showed that people with genotype rs6843082 (GG), with or without hypertension had OSA as a risk factor for stroke. For individuals with the genotype rs6843082 (GA + AA), those with hypertension, OSA is a risk factor for stroke, and for those without hypertension, OSA is not associated with stroke.

Data availability

The use of these data is restricted by regulations established by the Taiwan Biobank Dataset Authority and therefore the data are not publicly available. Data available on request due to privacy/ethical restrictions. The data that support the findings of this study are available on request from the corresponding author.

Received: 29 May 2024; Accepted: 30 September 2024

Published online: 23 October 2024

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Acknowledgements

The authors thank the National Science Council and the National Institute of Health for support. Data analyzed in this study were collected by the Taiwan Biobank and the National Health Insurance Research Database.

Author contributions

T.Y., Y.P.L. contributed to conception and design of the study. S.H. organized the database. S.H., Y.P.L. and D.T. performed the statistical analysis and methodology. T.Y. wrote the original draft of the manuscript. Y.C., H.H., J.P. and Y.C.L. review and editing sections of the manuscript. H.Y., Y.P.L. are Supervision. All authors contributed to manuscript revision, read, and approved the submitted version of the manuscript.

Funding

This work was funded by the Ministry of Science and Technology, Taiwan (MOST 109-2121-M-040-002; 109-2811-M-040-500; 110-2811-M-040-001; 110-2121-M-040-002; 111-2811-M-040-001; 111-2121-M-040-002; 111-2811-M-040-001). Sponsors had no role in study design, data collection, analysis, decision to publish or manuscript preparation.

Declarations

Competing interests

The authors declare no competing interests.

Ethical approval and consent to participate

The Institutional Review Board of Chung Shan Medical University Hospital (CS1-20009) approved this study. This study is not a clinical trial and it has been confirmed that all methods were conducted in accordance with relevant guidelines and regulations, including compliance with legal requirements in the country in which the study was conducted.

Consent for publication

All authors contributed to manuscript revision, read, and approved the submitted version of the manuscript.

Additional information

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1038/s41598-024-74782-x>.

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